

Toward a better understanding of curve number and initial abstraction ratio values from a large sample of watersheds perspective

Abderraman R. Amorim Brandão^{a,*}, Dimaghi Schwamback^{a,b}, André Simões Ballarin^a, John J. Ramirez-Avila^c, José Goes Vasconcelos Neto^e, Paulo Tarso S. Oliveira^{a,d,e}

^a Department of Hydraulics and Sanitation, São Carlos School of Engineering, University of São Paulo, CxP. 359, São Carlos 13566-590 São Paulo, Brazil

^b Division of Water Resources Engineering, Department of Building and Environmental Technology, Lund University, Box 118, Lund SE-221 00, Sweden

^c Watersheds and Water Quality Research Lab, Richard A. Rula School of Civil and Environmental Engineering, Mississippi State University, Mississippi State, MS 39762, USA

^d Faculty of Engineering, Architecture and Urbanism, and Geography, Federal University of Mato Grosso do Sul, Campo Grande 79070-900, Brazil

^e Department of Civil and Environmental Engineering, Auburn University, 238 Harbert Engineering Center, Auburn 36849, USA

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ABSTRACT

The Natural Resources Conservation Service Curve Number method (NRCS-CN) is the most widely used approach to estimate runoff from rainfall events. However, some uncertainties in the method remain linked to the value of the standard initial abstraction ratio (λ) and discrepancies between computed and standard tabulated Curve Number (CN) values. Here, we compute CN values and investigate the effects of λ on runoff estimation performance using a large sample of 3,578 watersheds distributed across the globe. We evaluate the impact of the default λ value of 0.2 and the proposed value of 0.05 across three methods and examine two rainfall event thresholds for CN calibration. We further investigated the association between the parameters of the NRCS-CN method and the catchment and climatic characteristics of the watersheds using machine learning techniques. Our findings indicated that the Least-Squares (LS) method better calibrates CN values and performs more accurately using $\lambda = 0.05$ compared to the default $\lambda = 0.2$ and that the 25.4 mm precipitation threshold showed better performance for calibrating the methods. The CN spatial variability revealed that high values of CN are controlled by the spatial variation of slope, precipitation, and soil silt content, while lower CNs aligned with forest lands, and strongly correlated to regions of sandy soils, down to the aridity index. Land cover emerges as the most influential characteristic in determining the λ , with cropland percentage exhibiting the greater influence. Arid regions, increases cropland, urban areas, and soil sand content are associated with $\lambda = 0.05$, whereas higher pasture percentages correspond to $\lambda = 0.2$. We also provide equations for converting parameters from $\lambda = 0.2$ to $\lambda = 0.05$. Several watersheds worldwide are ungauged, and this study emphasizes the non-linear and complex nature of hydrological processes influencing the NRCS-CN method parameters. Our study provides a better understanding of the NRCS-CN method, bringing significant practical implications for various applications, including hydrological processes, stormwater management, flood forecasting, sediment management, hydrological modeling, and water resources engineering.

1. Introduction

Precise estimations of runoff are crucial for a better understanding of hydrological processes, for the design of hydraulic infrastructure, and for planning and managing water resources. Hydrologists have dedicated considerable efforts to develop different approaches to estimate this hydrological variable, especially because most watersheds around the world are not monitored (Eslami et al., 2023; Hrachowitz et al.,

2013). Hence, predicting runoff in ungauged basins (PUB) still represents a challenge in hydrology and water resources engineering (Guo et al., 2021).

Nowadays, there are various methods to estimate runoff from rainfall events in watersheds (Hrachowitz et al., 2013). Among them, the Natural Resources Conservation Service (NRCS) developed a practical and widely used empirical approach known as the NRCS Curve Number method (NRCS-CN) (USDA-NRCS, 1986; Yu, 1998). This method relies

* Corresponding author.

E-mail address: abderraman.brandao@usp.br (A.R.A. Brandão).

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on the use of tables describing runoff susceptibility, considering hydrologic conditions, land use, and soil types (Bartlett et al., 2016). The prominence of the NRCS-CN method among hydrological models (Techapinyawat et al., 2023; Verma et al., 2017), such as Soil and Water Assessment Tool (SWAT) (Arnold et al., 1998), Storm Water Management Model (SWMM) (U.S. Environmental Protection Agency, 2022), and Hydrologic Engineering Center – Hydrologic Modeling System (HEC-HMS) (U.S. Army Corps of Engineers, 2023), underscores the importance of this method, highlighting the imperative need to enhance the method's accuracy. Such refinement is essential, as its widespread application spans diverse hydrological areas, such as flood control (Massari et al., 2023; Abd-Elaty et al., 2023), water quality (Leithold et al., 2023; Jin et al., 2023), sediment yield (Tadesse and Dai, 2019; Rostamian et al., 2008), runoff nutrient losses (Ramirez-Avila et al., 2017), and drainage design (Patil et al., 2023), among others.

One of the main advantages of the NRCS-CN method is its simplicity and availability of necessary data to compute runoff. The main factors driving surface runoff generation in this method – i.e. soil type, land cover and land use, surface condition, and antecedent runoff condition – are combined into a single parameter, the CN value (Hawkins et al., 2009). NRCS established a standard CN values table originally developed for agricultural watersheds with a size of less than 250 km² in the United States, as detailed in the NRCS National Engineering Handbook Part 630 (NEH 630), chapter 9 (USDA-NRCS, 1986). The empirical nature of the established standard CN limited its applicability to broader environmental conditions and geographical regions (Ponce and Hawkins, 1996). However, the method has undergone transformations since the 1950s (Hawkins et al., 2020, 2009), and these changes have allowed its application in varied regions, including urban areas (Ponce and Hawkins, 1996). However, understanding the influence of watershed physical characteristics and hydrometeorological variables on the NRCS-CN method is still a subject of study in recent works (Simpson et al., 2023; Moglen et al., 2022; Krajewski et al., 2020). Particularly, these studies have focused on understanding the influence of rainfall on model performance (Simpson et al., 2023; Chin, 2022; Moglen et al., 2022; Lian et al., 2020), initial abstraction (Ia) and initial abstraction ratio (λ) on model performance (Ayers and Ramirez-Avila, 2024; Lasco and Moglen, 2023; Junior Valle et al., 2019; Woodward et al., 2003), and computing CN values by using rainfall and runoff data (Ramirez-Avila et al., 2024; Sharma et al., 2022; Lian et al., 2020; Durán-Barroso et al., 2017; Oliveira et al., 2016; D'Asaro et al., 2014). In general, these studies have revealed the need for i. a better understanding of natural and ordered rainfall-runoff data and the threshold of rainfall to estimate CN; ii. the potential use of $\lambda = 0.05$ instead of 0.20; and iii. discrepancies between computed and standard tabulated CN values, highlighting the need for local/regional estimates of CN values.

The standard tabulated CN values are derived from the median of the values obtained from rainfall-runoff pairs (Hawkins et al., 2009). However, since their first appearance, researchers have explored alternative methods to determine CN values, highlighting the asymptotic (ASY) (Hawkins, 1993) and nonlinear Least-Squares (LS) methods (Hawkins, 1993). Their findings, however, are highly variable and the definition of the most suitable method is subject to an ongoing scientific debate. Some researchers found that ASY better determines the representative CN value for a drainage area (Galbetti et al., 2022; Lian et al., 2020; Oliveira et al., 2016; D'Asaro et al., 2014). Other studies suggested that the LS method exhibits a superior performance (Ramirez-Avila et al., 2024; Pezan, 2022; Moglen et al., 2022; Ling et al., 2020) or indicated that the traditional median (NEH) method is unable to accurately represent CN values (Ayers and Ramirez-Avila, 2024; Moglen et al., 2022; D'Asaro and Grillone, 2012). This considerable variability demonstrates a lack of consensus among researcher's findings regarding the most suitable method.

The generation of runoff in the NRCS-CN method occurs when precipitation exceeds a percentage of the storage capacity threshold, which is composed of interception, evaporation, infiltration, and surface

storage before runoff begins (Mishra et al., 2018; USDA-NRCS, 2004). The sum of these factors is usually referred as initial abstraction (Ia), which can be estimated as the multiplication of the λ by the storage capacity (S). Therefore, the Ia significantly influences runoff estimation (Yuan et al., 2014). When developing the method, the λ was set at 20%, but different studies have shown that the use of $\lambda = 0.20$ could considerably overestimate runoff (Verma et al., 2017; Woodward et al., 2003; Aron et al., 1977), and that this value should be smaller, despite values ranging from 0 to 0.99 depending on watershed characteristics (USDA-NRCS, 1986; Jiang, 2001), often converging to values of 0, 0.01, 0.03 and medians around 0.05 (Ramirez-Avila et al., 2024; Simpson et al., 2023; Durán-Barroso et al., 2017; Junior Valle et al., 2019; Hawkins et al., 2009; D'Asaro et al., 2014; Fu et al., 2011; Shi et al., 2009; Woodward et al., 2003; Jiang 2001). The CN values from table determined by the NEH method using $\lambda = 0.20$, lead to oversized engineering structures (Galbetti et al., 2022; Lal et al., 2017; Tedela et al., 2012) and more expensive engineering projects (Moglen et al., 2022; Tedela et al., 2012; Schneider and McCuen 2005). The ASCE-EWRI CN Hydrology Committee based on literature led to a consensus to consider the revised standard value of 0.05, a potential change currently under review in the USDA-NRCS draft document (Hawkins et al., 2020; USDA, 2017; ASCE-ASABE-NRCS Task Group on Curve Number Hydrology, 2017).

Research conducted in different regional studies has discussed optimal values for different watersheds. Hawkins et al. (2009), when analyzing λ values for 307 watersheds, found that for 82.1% of them, a ratio of 0.05 exhibited better performance in predicting runoff. Lasco and Moglen (2023), studying 47 watersheds, observed only slight differences and no statistically significant distinction between 0.2 and 0.05. Similarly, Moglen et al., (2022), in a study using 31 watersheds, reported relatively similar predictions between these indices. Conversely, global studies (e.g., Vamshi et al., 2023; Wu et al., 2023; Sujud and Jaafar, 2022; Jaafar et al., 2019) have utilized remote sensing techniques to obtain traditional tabulated CN values that assume a standard $\lambda = 0.2$.

In general, regional studies have focused on analyzing the method performance on runoff estimation for the studied watersheds varying the value of λ , without considering how the different physical and environmental watershed characteristics influence the method. These studies have also proposed conversion equations for previously tabulated CN values in TR-55 from CN_{0.2} to CN_{0.05} based on these small samples. In contrast, global studies have predominantly used the standard $\lambda = 0.2$ without such variations. Therefore, there is still an open scientific question on how physical and environmental watershed characteristics drive Ia and λ values. In other words, no study has assessed the watershed characteristics that align with $\lambda = 0.05$ or $\lambda = 0.2$ for more accurate runoff estimations. Furthermore, to the best of our knowledge, there is no study computing CN values from rainfall-runoff data using varying λ values for multiple countries, and investigating the method performance for runoff estimation, which would lead to more accurate conversion equations given the large number of watersheds with numerous characteristics.

The present study aims to fill this scientific gap by computing the CN values and investigating the effects of λ on the performance of the NRCS-CN on runoff estimation for 3578 watersheds distributed across the globe. We further investigate the main relationships of watershed characteristics with λ and CN values. We compared the performance of the default λ value of 0.2 against the proposed 0.05 using three different methods for computing CN (ASY, LS, and NEH) and two different groups of rainfall events: considering only rainfall events greater than 25.4 mm or all rainfall events. To identify how the physical and environmental watershed characteristics drivers λ and CN values, we use traditional statistics and machine learning methods. Lastly, we provide regression equations to convert S_{0.2} to S_{0.05}, and consequently CN_{0.05}, for the entire dataset and separately for each of the evaluated countries. Our findings may be useful for future applications of the NRCS-CN method

considering $\lambda = 0.05$, as it may be used to update the previously tabled CN values in TR-55. Hence, here we present the first comprehensive NRCS-CN method investigation using daily rainfall and runoff data across five continents and directs future analyses by offering a road map for the use of suitable CN and λ values.

2. Material and methods

This paper summarizes a complex workflow, presenting, in addition to the discussion of the optimal value of λ for the NRCS-CN method, the best method to find the CN value, the influence of precipitation events on calibration, and spatial analyses of CN values. Additionally, the study goes beyond traditional analyses, employing principal component analysis (PCA), random forest analysis (RFA), and SHapley Additive exPlanations (SHAP) for analysis and interpretation of the relationship between method parameters and watershed characteristics using from a large sample dataset, as summarized in the flow chart presented in Fig. 1. The methodology described in the following subsections is as follows: First, we present the data used in the study. Next, we explain the different calibration methods. Then, we describe the methods used to assess the relationship between watershed attributes and λ .

2.1. Dataset and data preparation

We used daily precipitation-runoff data accounting for 13,243 daily events spanning nearly seven decades (1950–2023), and physiographic attributes of 3674 watersheds with areas smaller than 250 km² (Ponce and Hawkins, 1996) distributed across five continents and 32 countries (Fig. 2). We used the Caravan dataset (Kratzert et al., 2023), with additional extensions of Denmark (Caravan-DK) (Koch and Schneider, 2022); the Global Runoff Data Centre (GRDC) (Färber et al., 2023), Israel (Efrat, 2023), Switzerland (CAMELS-CH) (Höge et al., 2023), Spain (CAMELS-ES) (Rodríguez, 2023) and Iceland (LamaH-Ice) (Helgason and Nijssen, 2023). The validation of the Caravan dataset involved statistical consistency analyses, comparing it with other established datasets; for more details on their approach to avoiding data errors, refer to Kratzert et al. (2023). For the analysis, we excluded days when runoff exceeded precipitation to avoid human interference effects (e.g., gate openings), and days with precipitation without a corresponding runoff record (Moglen et al., 2022), or events that did not generate runoff (Moglen et al., 2022; Hawkins et al., 2015a).

It is a common practice to calibrate the CN using only larger precipitation events (Ajmal et al., 2023; Moglen et al., 2022; Pezan, 2022; Wilson et al., 2017; Oliveira et al., 2016; Hawkins et al., 2009, 1985). However, the definition of what constitutes a “large” precipitation event remains ambiguous and the CN method should not be limited to only large events (Chin, 2023a; Durán-Barroso et al., 2017). According to Durán-Barroso et al. (2017), the commonly adopted threshold of 25.4 mm used in these studies was initially set by Hawkins et al. (1985) based merely on professional judgment. Despite being open to the analyst’s discretion, this criterion has been applied in several studies without any further revision (Durán-Barroso et al., 2017). Moreover, given the variability of infiltration in watershed behavior during smaller precipitation events (Moglen et al., 2022; Durán-Barroso et al., 2017). Herein, we conducted the analysis using two different thresholds for precipitation following (Moglen et al., 2022; Chin, 2022; Hawkins et al., 2020): Group (1) includes 3398 watersheds with events exceeding 25.4 mm (Fig. 2d) and Group (2) includes 3578 watersheds with all precipitation events (Fig. 2c). The difference in the number of watersheds between Group (1) and Group (2) is due to the exclusion of 180 watersheds from Group (1) that did not have any precipitation events exceeding 25.4 mm. The sample size of the evaluated watersheds varied from a minimum of 31 to a maximum of 74 years, with more than 60 % of them exhibiting at least 40 years of data. (Fig. 2b). See Table S1 in the supporting information S1 for more details.

Here, we used the Eckhardt (2005) digital filter to estimate surface runoff given its good performance to partition streamflow components, and for not being contingent upon the watershed characteristics, which favors its broad applicability (Xie et al., 2020). A description of the filter is provided in S1.

2.2. Processing curve number calibration methods

The Curve Number method provides a simple and practical approach for predicting runoff from different types of watersheds. The runoff depth (Q , mm) is calculated as a function of precipitation (P , mm), with the λ , and Curve Number (CN_2)

$$Q = \begin{cases} 0, & \text{for } P \leq \lambda S_i \\ \frac{(P - \lambda S_i)^2}{P + (1 - \lambda)S_i}, & \text{for } P > \lambda S_i \end{cases} \quad (1)$$

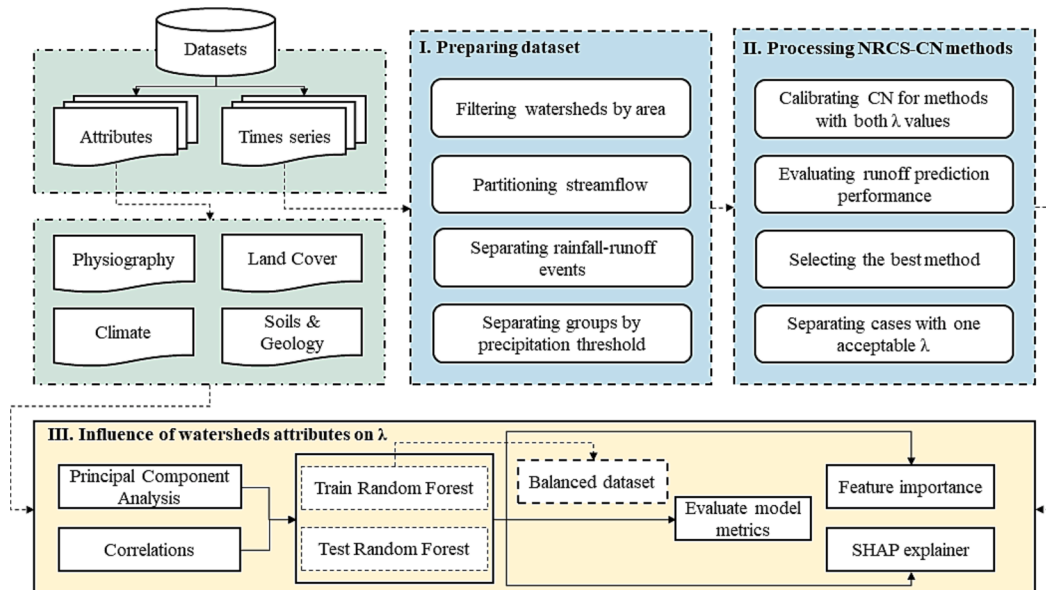


Fig. 1. Flow chart of the different steps followed in the study.

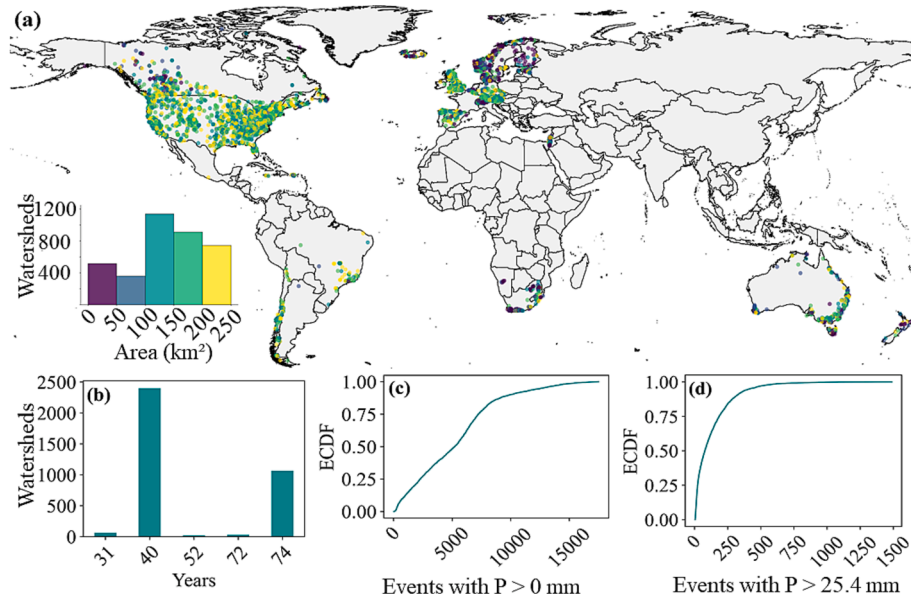


Fig. 2. Spatial distribution of the evaluated watersheds (a), frequency distribution of the dataset temporal length (b), cumulative distribution of the number of events (c), and cumulative distribution number of observations with $P > 25.4$ mm (d). Empirical Cumulative Distribution Function (ECDF) represents the proportion of observations less than or equal to a particular value.

$$CN_{\lambda} = \frac{25400}{254 + S_{\lambda}} \quad (2)$$

where (S_{λ} , mm) is the potential maximum retention, representing the storage capacity of the watershed for rainfall. As shown in Equation (1), λS_{λ} represents the initial abstraction (I_a), which corresponds to the amount of rainfall lost before runoff begins—where λ is the abstraction ratio corresponding to a specified λ value. Therefore, this study evaluates both $\lambda = 0.2$ and $\lambda = 0.05$, as they represent the standard and alternative values currently under consideration. CN_{λ} is a dimensionless parameter ranging from 0 to 100. A higher CN_{λ} indicates a greater potential for runoff generation in the watershed, with typical values ranging between 40 and 98 (Van Mullem, 1991). The potential maximum retention after runoff begins is influenced by precipitation and runoff. The value of S_{λ} can range from 0 to infinity, indicating the range of possible retention values depending on the specific rainfall and runoff features.

To compute the CN_{λ} values, we investigated three methods: the National Engineering Handbook median (NEH) (USDA-NRCS, 2004), the Least-Squares (LS) (Hawkins, 1993), and the Asymptotic (ASY) methods (Hawkins, 1993). The use of median and arithmetic mean is characterized by Schneider and Mccuen (2005) as the best method for estimating CN using storm events. The NEH method involves applying the median values of $S_{0.2}$ and $S_{0.05}$ to calibrate CN_{λ} . The equations for calculating S_{λ} , considering $\lambda = 0.2$ and $\lambda = 0.05$, are as proposed by Hawkins (1973) and Woodward et al. (2003):

$$S_{0.2} = 5 \left(P + 2Q - \sqrt{4Q^2 + PQ} \right) \text{ for } 0 \leq Q \leq P \quad (3)$$

$$S_{0.05} = 10 \left(2P + 19Q - \sqrt{361Q^2 + 80Q} \right) \text{ for } 0 \leq Q \leq P \quad (4)$$

The Least-Squares calibration of the Curve Number method involves finding the S_{λ} value that minimizes the observed and modeled runoff discrepancy (Equation (1), as defined an objective function Z:

$$z = \sum_{i=1}^n (Q_{obs,i} - Q_i)^2 \quad (5)$$

where $Q_{obs,i}$ represents the observed runoff generated from an event i

based on precipitation data, and Q_i represents the modeled runoff calculated using equation (1).

The Asymptotic approach for CN calibration is described by four different behaviors: standard, violent, complacent, and inactive (Hawkins et al., 2020; Hawkins, 1993). This method calculates the CN value using the rank-matched pairs of P and Q , considering that higher rainfall events will likely result in greater runoff (Hawkins et al., 2009). In this study, we follow the original approach by Hawkins (Hawkins, 1993), who formulated the method in 1993 using rank-matched pairs, and some authors have found that using rank-matched pairs yields better results compared to natural pairs (Moglen et al., 2022; Hawkins et al., 2009). For the standard behavior, the method relates the decline of precipitation and CN to calibrate the relationship between the two variables, given by the equation:

$$CN_{\lambda}(P) = CN_{\infty} + (100 - CN_{\infty})e^{-k_1 P} \quad (6)$$

where CN_{∞} is the asymptotic CN value, and k_1 is the adjustment coefficient obtained through a least-squares procedure. This method places more weight on larger precipitation events in the dataset (Moglen et al., 2022). Similarly, the violent behavior adjusts to an exponential function, with CN values rapidly increasing with precipitation and approaching a high constant value as they trend toward CN_{∞} . The equation for violent behavior, with k_2 as a distinct adjustment constant, is:

$$CN_{\lambda}(P) = CN_{\infty}(1 - e^{-k_2 P}) \quad (7)$$

The Complacent behavior does not yield a definitive CN value (Verma et al., 2017), and in such cases, a linear relationship between Q and P , which slope represents the runoff coefficient (C), is more descriptive (Sartori et al., 2011; Hawkins et al., 2009). This is often the case due to the high infiltration capacities that could be found in many upland forest soils (Hawkins et al., 2020). Lastly, the inactive behavior is evaluated where there is no defined condition. Despite the separation into these behaviors, their definition is based on a visual adjustment. The standard behavior accounts for the majority of datasets, being compatible with CN procedures (Hawkins et al., 2020, 2009) in over 80% of the cases (Hawkins et al., 2020). The complacent behavior can be approximated to equation (6) but with uncertainty for larger precipitation events (D'Asaro and Grillone, 2012b).

The generation of runoff in a drainage area is also influenced by the antecedent runoff conditions (ARC). Three classes of ARC (I, II, and III) have been defined and were initially determined for dormant and growing seasons, based on the average precipitation over the five days preceding an event. However, this criterion did not accurately represent the variability across different regions and was no longer used (Hawkins et al., 2009). Given the regional variability of ARC, the ARC II class has been reaffirmed as an appropriate basis for design procedures, as it accurately portrays all three conditions. Tabulated CN values represent ARC II conditions (CN_{II}) (Hawkins, 2010; Hawkins et al., 2009). The CN values reported in this study are defined to represent the ARC II, adjusting the standard behavior of runoff.

2.3. Influence of watershed attributes on λ

We investigated the relationship between physical and environmental watershed characteristics and the parameters $\lambda = 0.05$ and $\lambda = 0.2$ by using correlation analyses, Principal Component Analysis (PCA), Random Forest Model (RFM), and SHapley Additive exPlanations (SHAP), considering both balanced and unbalanced datasets (i.e., balanced datasets have an equal number of representations across classes or categories, while unbalanced datasets do not). Here, we investigated the association between watershed attributes and λ for watersheds where the NRCS-CN method provided the better performance in runoff estimation with a Kling-Gupta efficiency (KGE) (Gupta et al., 2009) value above 0.5. This metric unifies bias, correlation, and variability, offering a solution to the limitations of the Nash Sutcliffe Efficiency (NSE) (Chadalawada and Babovic, 2019). Most authors generally describe this threshold (KGE = 0.5) as an acceptable limit (Ajmal et al., 2023; Pezan, 2022; Towner et al., 2019).

2.3.1. Interpretation through principal component analysis (PCA)

A preliminary analysis eliminated redundant variables, such as nighttime lights and population density, which do not influence the method. PCA was applied to a subset of 12 attributes among the more than 200 attributes listed in the CARAVAN dataset, considering both catchment watershed and climate characteristics (percentage of forest cover, cropland area, pasture area, urban area, clay content, silt content, sand content, aridity index (the ratio between mean-annual potential evapotranspiration and precipitation derived from ERA5-Land (Muñoz Sabater et al., 2021), average daily precipitation, watershed area, high rainfall days ($\geq 5 \times$ times daily mean), and upland slope) (see Table S1 in S1).

The application of PCA aims to reduce the dimensionality of the input data series while preserving its variance. This transformation reprojects the data into a smaller set of orthogonal, linearly independent, and uncorrelated variables known as principal components (PCs) (Nourali, 2023; Bickel et al., 2009). The variance of each principal component is given by the eigenvalue associated with it. The first principal component (PC1) is the linear combination with the largest variance, with each subsequent PC having the next largest variance, subject to being orthogonal to the preceding component. The total variance explained if preserved by the PC is the sum of variances from each PC. PCA aids in data interpretation by eliminating multi-collinear variables. It ensures that the first PC captures the most variability in the data and that data points are discriminative in any given direction.

2.3.2. Random forest analysis

To identify which λ value is more suitable for a particular watershed based on the attributes found from PCA, we used the Random Forest classifier. It is a popular machine-learning algorithm found in various packages due to its simplicity of training and tuning, and its performance is very similar to boosting (Bickel et al., 2009). It is an ensemble of a large collection of de-correlated trees built using bootstrap, and, as a decision tree ensemble, it is subject to reduced overfitting (Breiman, 2001).

The RFM offers various mechanisms for assessing the importance of an input variable, thereby enhancing model interpretation. The variable importance is determined by how much it improves the data separation in each division where it is used, averaged across all trees in the forest (Bickel et al., 2009). Breiman (2001) proposed a measure to evaluate the variable importance $\text{Imp}(X_j)$, based on the contribution of a variable X_j in the prediction of the target variable Y (Equation (8)). This measure considers the decreases in impurity $p(t)\Delta i(s_t, t)$ for all nodes t where X_j is used for splitting, averaged over all trees ϕ_m (for $m = 1, \dots, M$)

$$\text{Imp}(X_j) = \frac{1}{M} \sum_{m=1}^M \sum_{t \in \phi_m} [1(j_t = j)p(t)\Delta i(s_t, t)] \quad (8)$$

where, M is the total number of trees in the random forest; ϕ_m is the set of nodes in a tree, m , $p(t)$ represents the proportion of samples reaching a node; and $t \in \Delta i(s_t, t)$ denotes the decrease in impurity resulting from the split. The term j_t indicates the identifier of the variable used to divide a node t . The impurity function $i(t)$ can be any (e.g. Gini or Entropy) and the termed mean decrease impurity and represents the information gained (Louppe, 2014). The Random Forest model, its complexity and set of equations are further explored in (Breiman, 2001).

In this study, the RFM is constructed using Scikit-learn. For these parameters, the GridSearch function is utilized to find the optimal settings: (i) the number of estimators, a range of 100–500; (ii) max features, the square root of the number of attributes; (iii) for a minimum sample and number of samples split of 10, 20, 30 leaves and other parameters are set to their default values. Additionally, 20% of the data is used for testing, and a stratified ten-fold cross-validation is employed to prevent over-fitting or under-fitting during model fitting.

2.3.3. SHapley Additive exPlanations (SHAP) for variable analysis

RFM provides the importance of the variables but does not show how the relationship of their increase or decrease affects the target. To address this limitation, we used the SHAP approach, type of explainable artificial intelligence (XAI), based on game theory (i.e., it suggests that each player contributes uniquely to the overall outcome of the game) (Lundberg and Lee, 2017). This approach is intuitive for human understanding and offers the perspective that any explanation for the prediction of a model can be seen as a model itself, described by Equation (9):

$$\omega_i(f, x) = \sum_{T \subseteq F \setminus \{i\}} \frac{|T|!(|F| - |T| - 1)!}{|F|!} [f_{T \cup \{i\}}(x_{T \cup \{i\}}) - f_T(x_T)] \quad (9)$$

where, ω_i is the SHAP value for feature i . The summation runs over all subsets T of features, excluding feature i . $|T|$ is the number of features in T , and $|F|$ is the total number of features. $f_{T \cup \{i\}}(x_{T \cup \{i\}})$ is the model's predicted value with the features in T plus feature i , and $f_T(x_T)$ is the predicted value of the model with just the features in T . This equation allows for the calculation of the marginal contribution of feature i his equation allows for the calculation of the marginal contribution of feature i . SHAP overcomes the dependency on using any specific model and can be used to align explanations from different models, as it is model-independent (Dahal et al., 2023). SHAP's complexity is not summarized just in the Equation (9), for this set of equations is further explored in Chen et al. (2022) and Lundberg and Lee (2017).

2.3.4. Imbalanced in machine learning

Machine learning algorithms are often influenced by a class imbalance in the target dataset (Batista et al., 2004). This imbalance can lead to model bias, favoring the majority class and impairing the classification accuracy of minority classes (Batista et al., 2004). In this context, the Synthetic Minority Over-sampling Technique (SMOTE), proposed by (Chawla et al., 2002), is a common technique to mitigate this overfitting problem (Batista et al., 2004). SMOTE involves creating new examples of the minority class by forming them along line segments connecting

minority class samples with their nearest neighbors, also belonging to that class (Chawla et al., 2002). While classic SMOTE focuses exclusively on oversampling the minority class, SMOTE-Tomek combines this with the elimination of examples from the majority class that are in excessive proximity to the minority class (Batista et al., 2004). This process is facilitated by the use of the k-nearest neighbors (k-NN) algorithms to identify and remove such potentially problematic examples. The main advantage of SMOTE-Tomek over standard SMOTE lies in its ability to reduce the overlap between classes, enhancing the creation of a clearer decision boundary. Given the possibility of imbalance, two datasets are utilized: the original and the balanced.

2.4. Performance evaluation

The performance of each method used to determine the CN values, considering the estimated and observed surface runoff, was based on the determination of the Coefficient of Determination (R^2); Equation (10), Relative Standard Error (RSE; Equation (11), Nash-Sutcliffe Efficiency Coefficient (NSE; (Nash and Sutcliffe, 1970); Equation (12), Percentage Bias (PBIAS; Equation (13), and Kling-Gupta Efficiency (KGE; (Gupta et al., 2009); Equation (14).

$$R^2 = \left[\frac{\sum_{i=1}^n (y_{i,obs} - \bar{y}_{i,obs}) \sum_{i=1}^n (y_{i,sim} - \bar{y}_{i,sim})}{\sqrt{\sum_{i=1}^n (y_{i,obs} - \bar{y}_{i,obs})^2 \sum_{i=1}^n (y_{i,sim} - \bar{y}_{i,sim})^2}} \right]^2 \quad 0 \leq R^2 \leq 1.0 \quad (10)$$

$$RSE = \sqrt{\frac{n \sum_{i=1}^n (y_{i,obs} - y_{i,sim})^2}{(n-2) \sum_{i=1}^n (y_{i,obs} - \bar{y}_{i,obs})^2}} \quad 0 < RSE < \infty \quad (11)$$

$$NSE = 1 - \frac{\sum_{i=1}^n (y_{i,sim} - y_{i,obs})^2}{\sum_{i=1}^n (y_{i,obs} - \bar{y}_{i,obs})^2} \quad 0 < NSE \leq 1.0 \quad (12)$$

$$PBIAS = \frac{\sum_{i=1}^n (y_{i,obs} - y_{i,sim})}{\sum_{i=1}^n y_{i,obs}} \times 100 \quad -\infty < PBIAS < \infty \quad (13)$$

$$KGE = 1 - \sqrt{(r-1)^2 + (\beta-1)^2 + (\gamma-1)^2} \quad -\infty < PBIAS \leq 1 \quad (14)$$

wherein: n is the sample size, $y_{i,sim}$ and $y_{i,obs}$ are the predicted and observed data, respectively, $\bar{y}_{i,obs}$ is the mean value of the observed data, r linear correlation between observations and simulations, β represents the bias, and γ represents the standard deviation ratio of the simulated

and observed data, respectively. PBIAS is given in percentage, and R^2 , RSE, NSE, and KGE are dimensionless indices. In the performance evaluation of classification models, assessments were carried out through the analysis metrics Accuracy, Precision, Recall, Specificity, Receiver Operating Characteristic (ROC) values and Area Under the Curve (AUC). For both precipitation groups, where at least one of the indices is satisfactory, we analyzed the results in series with less than 50 data points using the Shapiro test (Shapiro and Wilk, 1965), and for larger series, the Kolmogorov-Smirnov test (Razali and Wah, 2011), to assess normality, followed by the Wilcoxon test (Wilcoxon, 1945) to evaluate significant differences between the observed and estimated runoff for $\lambda = 0.05$ and $\lambda = 0.2$.

3. Results

3.1. Comparative assessment of CN determination methods

Fig. 3 shows the results obtained by calibrating the CN value using ASY, LS, and NEH, for two λ values and two precipitation assessment groups. Median CN values of 62.68, 64.07, and 69.48 for the ASY, LS, and NEH methods, respectively, were calibrated with $\lambda = 0.2$ for Group (1). In contrast, median CN values of 57.45, 64.47, and 97.75 were determined for Group (2). Regarding $\lambda = 0.05$, the median CN values for each corresponding method were 44.96, 44.65, and 43.73 for Group (1), and 52.69, 46.87, and 94.56 for Group (2). The dispersion of the CN values denoted by the range of the boxes and the whiskers of the box-plots has notable variability for the ASY and LS methods, particularly for Group (2), while the NEH method shows higher CN values and lower variability.

NEH calibrations generated the most significant difference between CN medians values of Group (2) and Group (1). These values increased 28.92% with $\lambda = 0.2$ and 53.75% with $\lambda = 0.05$, evidencing that NEH is highly sensitive to the precipitation event threshold defined to determine CN. For the remaining methods, the difference is subtle: using ASY the CN value decreased by 9.10% with $\lambda = 0.2$ and 14.67% with $\lambda = 0.05$ when $P > 0$ mm. Conversely, using LS, the CN value increased by 0.08% with $\lambda = 0.2$ and 4.73% with $\lambda = 0.05$, indicating less sensitivity to the magnitude of precipitation events than the other methods. For all three methods, the CN values found with $\lambda = 0.05$ were lower than those with $\lambda = 0.2$. This behavior tends to happen mainly for Group (1) events. Figs. S1 to S12 in S2 represent the distribution of CN values across different countries for each method and Group.

CN values determined by the ASY, LS and NEH methods varied across land covers, particularly in watersheds where this use predominates (76 to 100%). For instance, using the LS method, in forest lands, the average

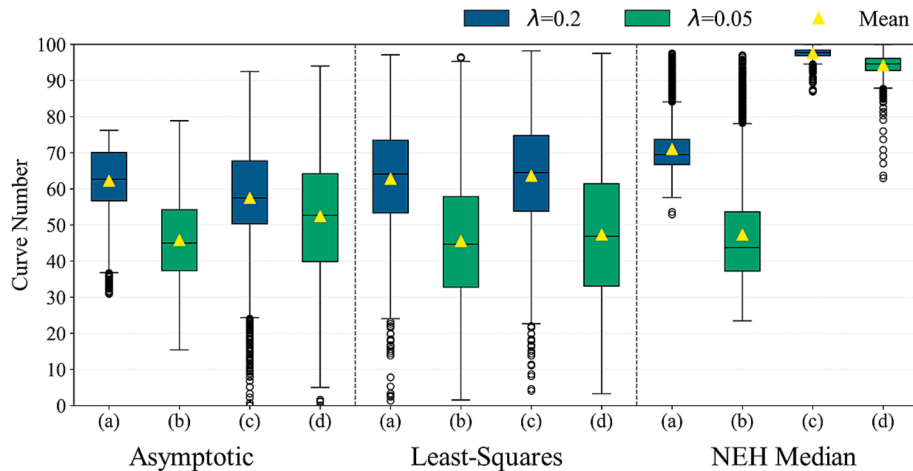


Fig. 3. Distribution of CN values for watersheds determined by three methods across four scenarios: (a) events with precipitation $P > 25.4$ mm, initial abstraction ratio (λ) of 0.2; (b) events with $P > 25.4$ mm, λ of 0.05; (c) events with $P > 0$ mm, λ of 0.2; and (d) events with $P > 0$ mm, λ of 0.05.

CN values ranging from 46.49 ($\lambda = 0.05$) to 63.02 ($\lambda = 0.2$) in Group (1), and 47.79 ($\lambda = 0.05$) to 63.04 ($\lambda = 0.2$) in Group (2). In croplands, CN ranged from 37.24 ($\lambda = 0.05$) to 57.82 ($\lambda = 0.2$) in Group (1), and from 39.05 ($\lambda = 0.05$) to 58.25 ($\lambda = 0.2$) in Group (2). For pasture lands, from 56.65 ($\lambda = 0.05$) to 70.99 ($\lambda = 0.2$) in Group (1), and from 55.43 ($\lambda = 0.05$) to 67.95 ($\lambda = 0.2$) in Group (2). Urban areas CN ranged from 56.24 ($\lambda = 0.05$) to 69.98 ($\lambda = 0.2$) in Group (1), and from 59.73 ($\lambda = 0.05$) to 71.16 ($\lambda = 0.2$) in Group (2). Across these land covers, CN values with $\lambda = 0.05$ were consistently lower than with $\lambda = 0.2$. Larger cropland areas tend to yield lower CN values while increasing pasture or urban areas results in higher CN values. Pasture lands showed a significant spread in CN values. For more detailed information about CN values for the method among different land covers, refer to S3 Figs. S13 to S18, and Tables S2 to S9.

Results from the PCA help disentangle the relationships between the CN spatial variability and watershed characteristics (refer to S4 Figs. S19 and S20). The analysis highlights hotspots where the spatial variability of CN values follows the PCs, revealing that the first component (PC1) was predominantly influenced by croplands, which have intermediate CN values, while the second component (PC2) was associated with the aridity index. The biplot clusters reveal that high CN values – which indicates hotspots for high runoff potential – were governed by the variability in slope, precipitation, and soil silt content. In contrast, low CN values – which indicates low runoff potential hotspots – were linked to forest land cover, areas strongly correlated to regions with sandy soils, extending towards the aridity index. This pattern is consistent across precipitation groups and became more discernible with $\lambda = 0.05$, which produces clearer CN clusters.

In relation to the performance of the three methods for estimating CN values, after calibrating CN values to the standard-type asymptotic function, R^2 values greater than 0.5 were estimated indicating a good fit between observations and predictions. These R^2 values were found in 38.52% ($\lambda = 0.2$) and 18.80% ($\lambda = 0.05$), of the cases for Group (1), and in 82.08% ($\lambda = 0.2$) and 98.46% ($\lambda = 0.05$) for Group (2). Therefore, watersheds that did not fit the standard-type behavior would fit into the complacent-violent-inactive behavior. A linear relation between runoff and rainfall best describes the complacent behavior, in turn, represented by the runoff coefficient (C). Values for this parameter ranged from (0.00–0.76) and (0.00–0.57) for Groups (1) and (2), respectively. Correlations between CN and C include, $CN_{0.2}$ vs. C with $r = 0.38$ ($p < 0.05$) and $CN_{0.05}$ vs. C with $r = 0.32$ ($p < 0.05$) for Group (1), as well as $CN_{0.2}$ vs. C with $r = 0.28$ ($p < 0.05$) and $CN_{0.05}$ vs. C with $r = 0.31$ ($p < 0.05$) for Group (2). Areas with a higher concentration of urban and pasture areas tended towards increasing the C values regardless of the group; with Group (2) displaying higher and more condensed values. In addition, with multiple land covers, C values became less consistent, as seen by the number of outliers observed when only one type of soil was predominant (For supporting information, refer to S5 Figs. S21 and S22).

The Empirical Cumulative Distribution Function (ECDF) of the performance metrics for the methods used to estimate CN values (considering precipitation groups and λ values) overview the methods' accuracy, sensitivity, and bias (Fig. 4). From the perspective of all metrics — R^2 , RSE, NSE, PBIAS, and KGE — it became evident that the CN values estimated with $\lambda = 0.05$ induced a better relationship between observed and predicted runoff for all methods, with LS showing a better performance according to the evaluated metrics, particularly for Group (2). This trend is reinforced by the R^2 analysis, where LS and ASY had better adjustment with $\lambda = 0.05$, showing steeper curves for both precipitation groups, indicating an improvement in precision. The RSE metric complements these findings, indicating a comparatively inferior performance by NEH, especially for Group (2), but with a notable improvement with $\lambda = 0.05$. This position reverses for Group (1), where the accuracy of all methods increases with this adjustment. Observations showed that not all watersheds presented a standard deviation in the predicted volume (Fig. 4), because they did not satisfy the condition $P >$

la (i.e., no runoff predicted), with not all curves reaching an ECDF of 100%. However, using $\lambda = 0.05$ resulted in a shift of the curves toward the optimal values of the metrics compared to $\lambda = 0.2$, indicating that $\lambda = 0.05$ was a more suitable value to predict runoff.

A predominant trend of negative NSE values was observed when evaluating the NEH performance, indicating this method failed to reproduce the mean of the observed data. With an ECDF close to 100%, most watersheds exhibited negative values for Group (2) using NEH, whereas nearly 50% of the watersheds exhibited negative NSE values for LS and ASY, and slightly higher with $\lambda = 0.05$. In contrast, for Group (1), the slope of the curves decreased indicating a broader distribution of errors across watersheds, while the sensitivity to precipitation was primarily noted using NEH (ECDF dropped around 50% into the range of negative values, with $\lambda = 0.05$ still maintaining curves with lesser inclination). Regarding PBIAS, there is a tendency for overestimation by LS and ASY for Group (2), which was slightly mitigated with $\lambda = 0.05$, while NEH overestimated runoff, particularly with $\lambda = 0.2$. All ECDF curves for Group (1) showed that regardless of the CN calibration method the NRCS-CN method overestimated runoff values; however, this bias was reduced when the precipitation threshold was 25.4 mm and $\lambda = 0.05$. Finally, the KGE summarizes three dimensions of the methods' performance — correlation, bias, and variation — indicating (a) improved overall performance with $\lambda = 0.05$ for all methods, (b) LS as the best method, and (c) censoring of precipitation events increasing the performance of the NEH method.

Table 1 summarizes the percentage of watersheds with satisfactory performance across the methods and according to λ and precipitation groups, focusing on thresholds for good performance (Table S10 to S12 in S7 show the data each region). The results, mirroring those depicted in Fig. 4, showed that the LS method outperforms the others (superior performance across all metrics), especially for Group (1). The ASY performance notably improves in Group (1), indicating its sensitivity to higher precipitation events. The NEH method, demonstrates the lowest performance, particularly in Group (2), with only minimal improvements at Group (1). This suggests that while the NEH method was the most affected by precipitation threshold changes, it consistently underperformed across most of the metrics. The LS method has a higher percentage of watersheds in which the CN model had good or acceptable performance predicting runoff. A precipitation threshold did not show much influence on the results.

Analyzing the LS results across the watersheds, we found that $\lambda = 0.05$ outperformed $\lambda = 0.2$ in most cases. Specifically, for each watershed in Group (1) and Group (2), $\lambda = 0.05$ demonstrated better performance compared to $\lambda = 0.2$ in R^2 (49.26% vs. 79.74%), RSE (79.81% vs. 85.24%), NSE (79.84% vs. 85.30%), PBIAS (87.14% vs. 96.73%), and KGE (59.12% vs. 92.96%) across the evaluated watersheds. Although KGE is a robust metric as it encompasses statistical performance aspects of other efficiency metrics (Chadalawada and Babovic, 2019), relying solely on it can dilute the importance of individual metrics. Therefore, we used a combination of metrics to ensure a comprehensive evaluation. Nonetheless, under the perspective of the KGE metric, $\lambda = 0.05$ was identified as the superior approach (Fig. 4). S6 Figs. S23 to S26 illustrate the KGE values distribution across the countries using LS.

Considering the shedding light on the differences between both groups, we evaluated whether their performance was statistically significant. The statistical analysis for both groups centered on the results of the LS method, considering cases where, in at least one of the initial abstraction ratios, the KGE showed satisfactory performance. Comparing the KGE values found with both λ , the normality test resulted in $p < 0.05$ and the Wilcoxon test revealed a statistically significant difference in KGEs for between λ ($p < 0.05$) for both groups. Similarly, comparisons of calibrated CN values were significantly different ($p < 0.05$) for both groups. Analyzing the observed runoff values with the simulated ones, 43 watersheds showed adherence to the normal distribution for Group (1). The Wilcoxon test was applied cross-sectionally, with 69.01% of the watersheds using $\lambda = 0.05$, exhibiting $p > 0.05$,

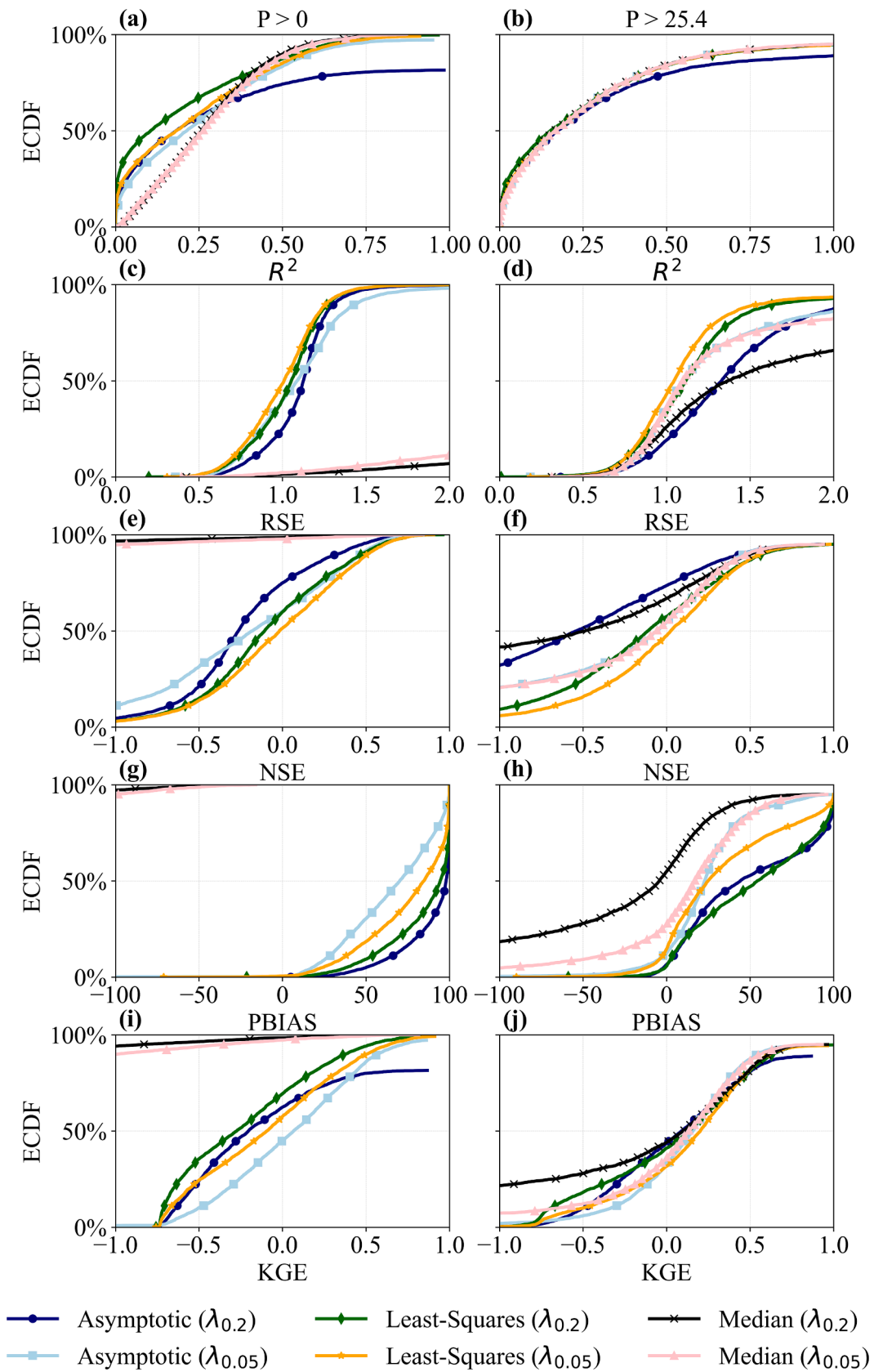


Fig. 4. The Empirical Cumulative Distribution Function (ECDFs) of evaluation model using ASY, LS, and NEH methods for $\lambda = 0.2$ and $\lambda = 0.05$. The first column representing conditions for precipitation $P > 0$ mm and the second column for $P > 25.4$ mm. Panels (a-b) show R^2 , (c-d) RSE, (e-f) NSE, (g-h) PBIAS, and (i-j) KGE distributions.

Table 1Summary of the percentage of watersheds with good fitting depending on the calibration method and λ .

Method	Percentage of watersheds ($\lambda = 0.05$)					Percentage of watersheds ($\lambda = 0.2$)				
	R ²	RSE	NSE	PBIAS	KGE	R ²	RSE	NSE	PBIAS	KGE
ASY _{group1}	11.39	38.31	4.00	54.76	7.65	9.86	20.16	3.91	39.20	8.50
ASY _{group2}	13.69	39.77	8.27	8.72	11.12	7.21	25.07	4.41	0.53	1.50
LS _{group1}	11.04	45.32	8.09	50.82	13.15	11.42	36.64	8.68	32.37	14.21
LS _{group2}	14.03	48.57	10.70	5.34	9.39	12.27	40.05	8.92	2.07	5.59
NEH _{group1}	11.54	37.85	4.62	49.47	8.80	11.39	27.16	5.44	48.06	13.50
NEH _{group2}	11.10	2.26	0.56	0.11	0.53	10.17	1.45	0.25	0.06	0.19

Note: the percentage of watersheds has indicated that the satisfactory results, for $NSE > 0.5$; $R^2 > 0.50$; $-25 \leq PBIAS \leq 25$ (Moriassi et al., 2007), $RSE < 1$ (Moglen et al., 2022), $KGE > 0.5$ (Ajmal et al., 2023; Pezan, 2022; Towner et al., 2019).

and for $\lambda = 0.2$, 59.13% had $p > 0.05$ when comparing runoff values. Furthermore, when comparing the surface runoff values within Group (1) across different λ values, 17.08% showed $p > 0.05$. In contrast, in Group (2), where no case followed the normal distribution, 100% of the cases showed significant differences ($p < 0.05$) in between observed and predicted values of runoff.

3.2. Conversion of CN from $\lambda = 0.2$ to $\lambda = 0.05$

The tabulated values of CN are based on $\lambda = 0.2$, and if there is a change to $\lambda = 0.05$ conversion equations to find new values of CN will be necessary as pointed out by Hawkins et al. 2009. Different authors (e.g., Woodward et al., 2003; Durán-Barroso et al., 2017; Junior Valle et al., 2019; Hawkins et al., 2020) have proposed conversion equations between $S_{0.05}$ and $S_{0.2}$. Using data from all watersheds, exponential and linear equations were fitted as displayed in Equations (17) and (18) for Group (1) and in Equations (19) and (20) for Group (2). Table 2 displays the same types of equations, considering each country separately to account for regional variations in hydrological response.

Table 2Conversions equations $S_{0.05}$ to $S_{0.2}$ (mm) for groups of watersheds.

Country	Group (1)						Group (2)					
	a	b	c	R ² (1)	R ² (2)	(n)	a	b	c	R ² (1)	R ² (2)	(n)
Argentina*	—	—	—	—	—	1	—	—	—	—	—	1
Australia	1.33	1.12	2.11	0.71	0.17	214	0.79	1.20	2.89	0.77	0.67	215
Austria	0.80	1.20	2.21	0.94	0.79	205	0.93	1.15	2.11	0.93	0.78	205
Belgium	1.34	1.10	2.19	0.91	0.83	24	1.67	1.00	1.70	0.90	0.64	25
Brazil	0.46	1.26	2.68	0.90	0.80	50	0.55	1.23	2.67	0.90	0.80	50
Canada	0.96	1.18	2.52	0.83	0.48	307	1.55	1.08	1.48	0.79	0.09	359
Chile	0.45	1.31	2.82	0.97	0.87	61	0.51	1.28	2.83	0.95	0.86	66
Czech Republic	0.30	1.40	2.94	0.90	0.79	14	0.18	1.50	2.87	0.88	0.75	14
Denmark	1.98	1.07	2.87	0.74	0.69	261	0.44	1.34	2.80	0.63	0.34	269
England	0.37	1.40	2.75	0.98	0.91	123	0.79	1.21	2.53	0.97	0.88	123
Estonia	0.24	1.49	3.41	0.90	0.87	16	0.22	1.49	3.33	0.95	0.89	24
Finland	1.36	1.11	2.71	0.81	0.79	27	0.70	1.21	2.50	0.74	0.71	28
France	3.00	0.90	1.56	0.93	0.65	18	3.21	0.86	1.43	0.97	0.82	18
Germany	0.73	1.22	2.37	0.91	0.58	130	0.95	1.15	2.12	0.93	0.78	131
Iceland	0.77	1.21	3.52	0.98	0.98	34	0.82	1.18	3.52	0.98	0.98	34
Ireland*	—	—	—	—	—	2	—	—	—	—	—	2
Israel	6.71	0.80	0.89	0.73	0.42	22	17.16	0.70	3.06	0.89	0.82	22
Italy	0.95	1.13	1.86	0.99	0.99	7	0.96	1.12	1.83	0.99	0.98	7
Jamaica	0.44	1.30	1.83	0.78	0.57	5	1.65	0.98	1.52	0.99	0.97	5
Luxembourg*	—	—	—	—	—	2	—	—	—	—	—	2
Mexico	—	—	—	—	—	1	—	—	—	—	—	1
Namibia	—	—	—	—	—	—	—	—	—	—	—	1
New Zealand	0.69	1.21	2.53	1.00	0.97	17	0.86	1.16	2.52	0.99	0.96	17
Norway	0.68	1.24	2.19	0.97	0.89	131	1.09	1.11	2.01	0.96	0.85	133
Puerto Rico	0.92	1.12	1.89	0.98	0.91	13	0.77	1.14	1.87	0.97	0.90	13
Scotland	0.73	1.21	1.69	1.00	0.95	64	0.99	1.16	1.74	0.99	0.98	64
South Africa	0.69	1.22	2.76	0.93	0.60	136	0.71	1.22	2.77	0.92	0.57	136
Spain	0.69	1.24	2.44	0.89	0.74	80	0.90	1.16	2.27	0.85	0.70	82
Sweden	0.70	1.26	2.93	0.91	0.80	39	0.40	1.36	2.84	0.88	0.75	39
Switzerland	0.85	1.18	2.15	0.96	0.87	111	1.07	1.11	2.04	0.95	0.85	111
U.S.A	1.17	1.11	1.00	0.87	0.52	1260	1.25	1.10	1.49	0.84	0.22	1358
Wales	0.60	1.26	1.74	1.00	0.94	23	1.06	1.13	1.70	0.99	0.99	23

Note: * Indicates insufficient data (n), $n \leq 3$. Equations follow: $S_{0.05} = a \cdot S_{0.2}^b$ (Equation (1)) and $S_{0.05} = c \cdot S_{0.2}$ (Equation (2)).

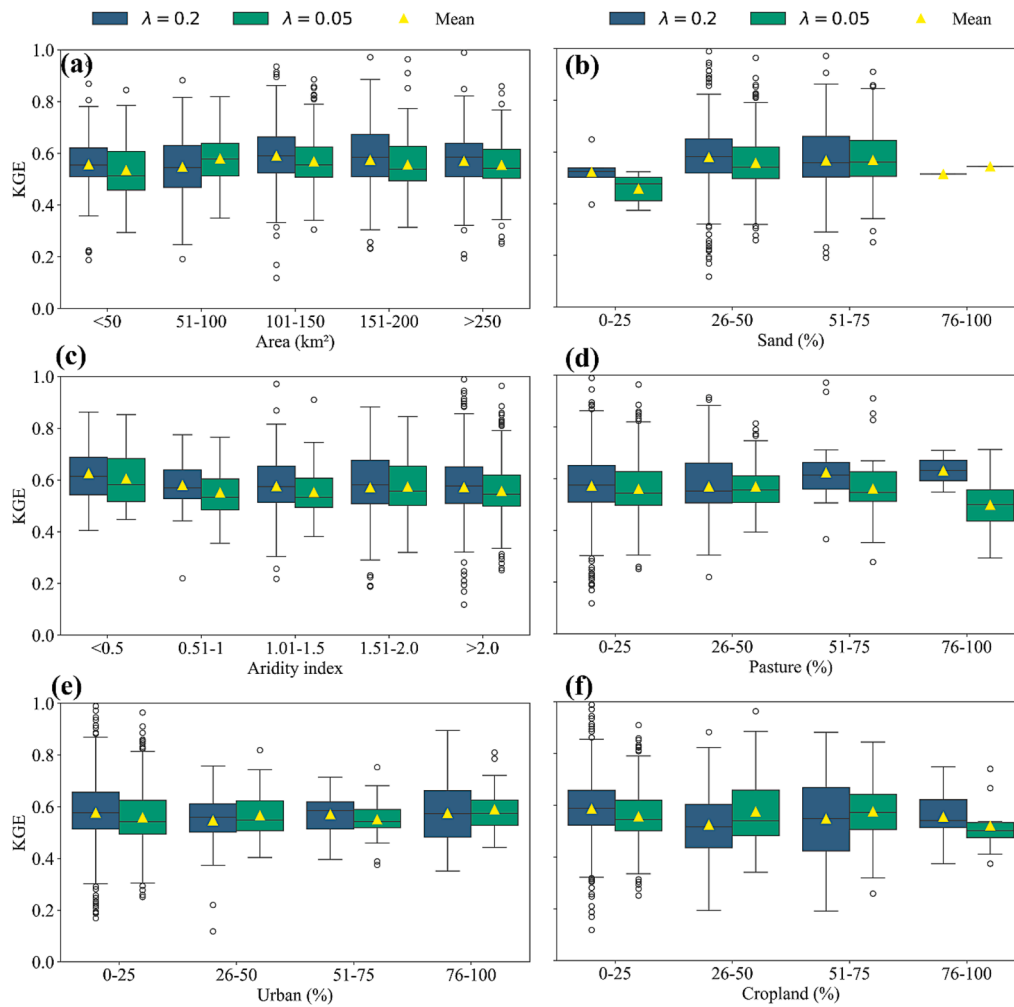


Fig. 5. Box-plot of the characteristics of watersheds that achieved a performance greater than 0.5 in KGE for $\lambda = 0.2$ or $\lambda = 0.05$ for precipitation events exceeding 25.4 mm.

While the medians were consistently close, the variability manifested by the interquartile ranges and outliers, particularly in the box plots for $\lambda = 0.2$, suggested the watershed response cannot entirely be attributed to the adjustment parameter, whereas for $\lambda = 0.05$ the box-plots were more concentrated with smaller whiskers. For Group (1), higher performance was found in highly urbanized watersheds (76–100%) with $\lambda = 0.05$, indicating that even at a lower precipitation an initial abstraction depth could be determined.

In contrast, a different pattern was observed for Group (2) (Fig. 6) with consistently higher KGE medians across all attributes' ranges when $\lambda = 0.05$, presenting more concentrated box-plots and smaller whiskers than when $\lambda = 0.2$. This phenomenon implies that a lower λ value facilitates capturing a broader range of runoff dynamics without penalizing the method's performance. For example, for Group (1) in watersheds with 0–25% sand content using $\lambda = 0.05$, a greater dispersion of KGE values and a lower median was observed, whereas for Group (2), the median increased and the dispersion decreased, suggesting an interaction between the composition of low sand content soils and low precipitation events associated to $\lambda = 0.05$. For watersheds with 76–100% sand content, a high KGE median of 0.60 was observed when $\lambda = 0.2$. Though based on a solitary watershed, it potentially relates a significant influence of soil composition on the effectiveness of hydrological simulation. However, given the limited sample size, these results warrant a cautious interpretation and highlight the scarcity of precipitation events greater than 25.4 mm in certain sandy regions.

For attributes ranges between 26–50% and 51–75%, a better

performance was predominantly observed when $\lambda = 0.05$ compared to 0.2 (KGE median = 0.60 for $\lambda = 0.05$ and 0.51 for $\lambda = 0.2$ for the first range, and 0.65 and 0.58, respectively for the second range). This was also evident for urban areas, where 85.84% of the watersheds within the 0–25% urbanization range had KGE medians of 0.63 and 0.53 when $\lambda = 0.05$ and $\lambda = 0.2$, respectively. Similar patterns were observed for 93.21% of the cropland and 71.39% of the pasture lands in the lower ranges. For watersheds between 101–150 km², the KGE values were higher when $\lambda = 0.05$ (median of 0.64) compared to $\lambda = 0.2$ (median of 0.54). For most of the watersheds with an aridity index greater than 2, higher KGE values were observed when $\lambda = 0.05$.

Results from the RFM (without applying SMOTE-Tomek) for Group (1) revealed a balanced distribution of the importance of variables for identifying the optimal λ that could better predict runoff. Watershed cropland (%) was the attribute that most importantly contributed (24.97%) to determining the optimal λ , while sand (%), aridity index, urban (%), pasture (%), and (drainage) area contributed 19.84%, 15.64%, 15.11%, 12.74%, and 11.68%, respectively. The application of SMOTE-Tomek in to RFM yielded a similar pattern, with contributions from cropland, urban, aridity index, sand, pasture and area, at 19.24%, 18.48%, 18.40%, 15.85%, 17.61%, and 10.40%, respectively. Performance metrics for the models were as follows: the ROC area was 0.70 and 0.71, respectively without and with SMOTE-Tomek. The corresponding Accuracy, Precision, Recall, and F1-scores were respectively 69.16%, 73.03%, 83.33%, and 77.84% without SMOTE-Tomek, and 65.83%, 77.61%, 66.66%, and 71.72% with SMOTE-Tomek, indicating

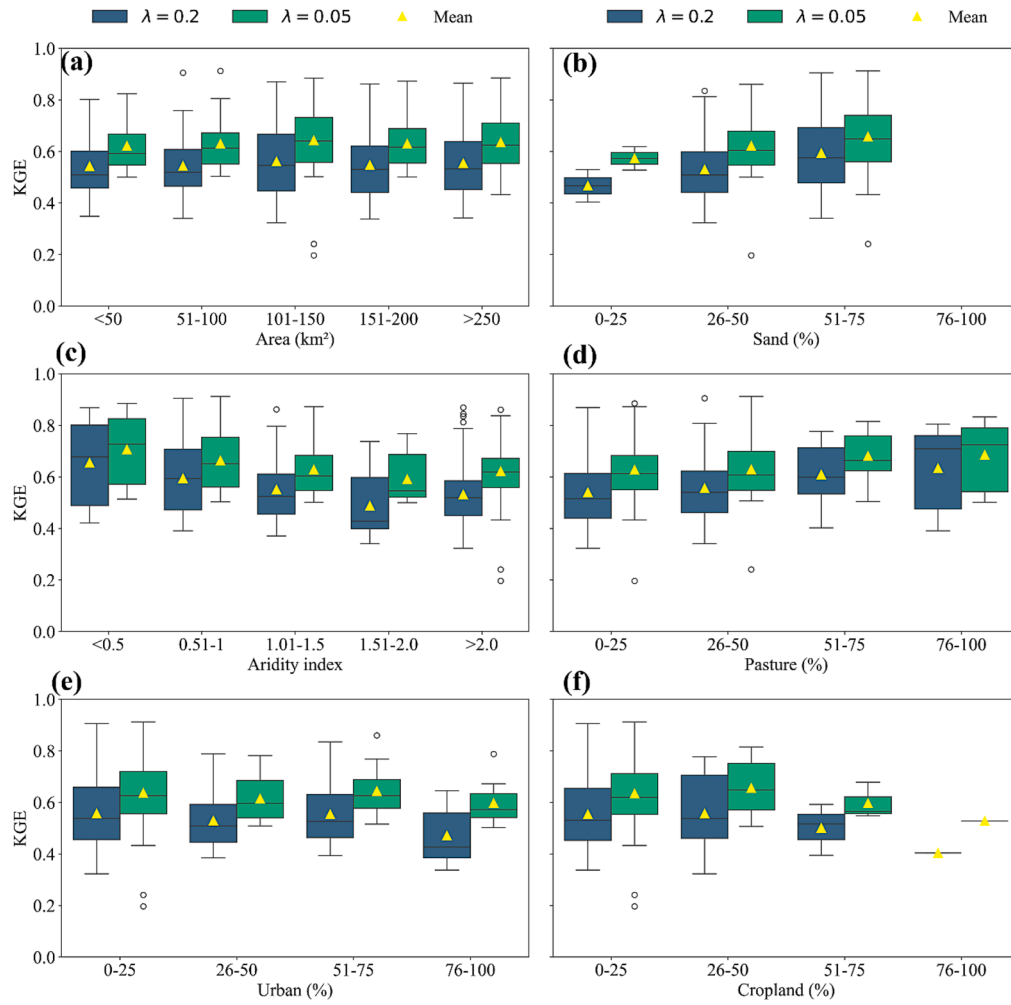


Fig. 6. Box-plot of the characteristics of watersheds that achieved a performance greater than 0.5 in KGE for $\lambda = 0.2$ or $\lambda = 0.05$ for all precipitation events.

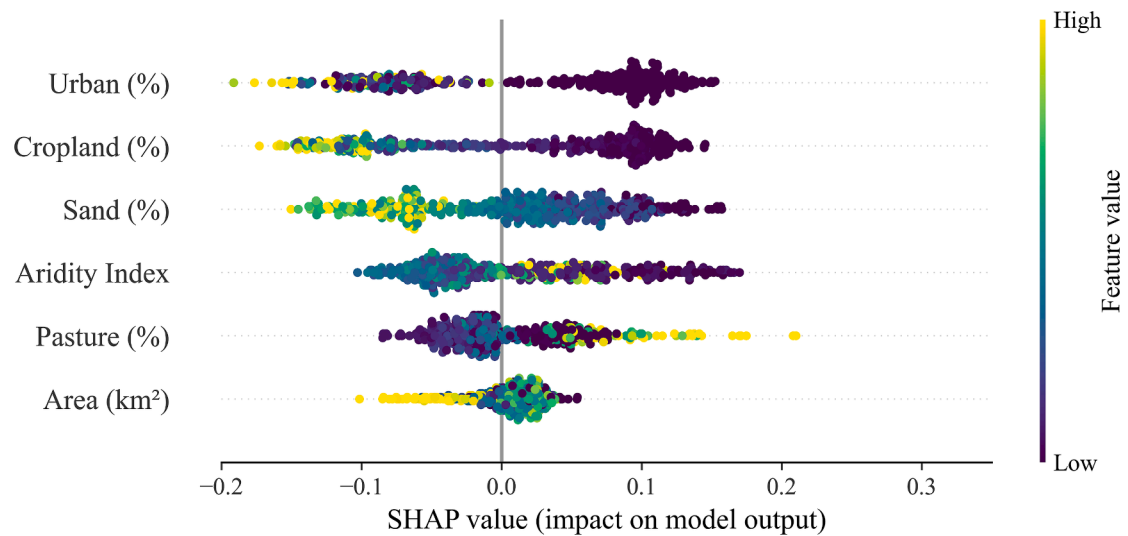


Fig. 7. Summary plot of SHAP values illustrating the impact of different features on the λ for Group (1) (positive class $\lambda = 0.2$) in balanced dataset. Each point on the plot represents the SHAP value for a feature for a single prediction, indicating the contribution of that feature to moving the model output from the base value. The color scale from green to purple represents feature values from high to low, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

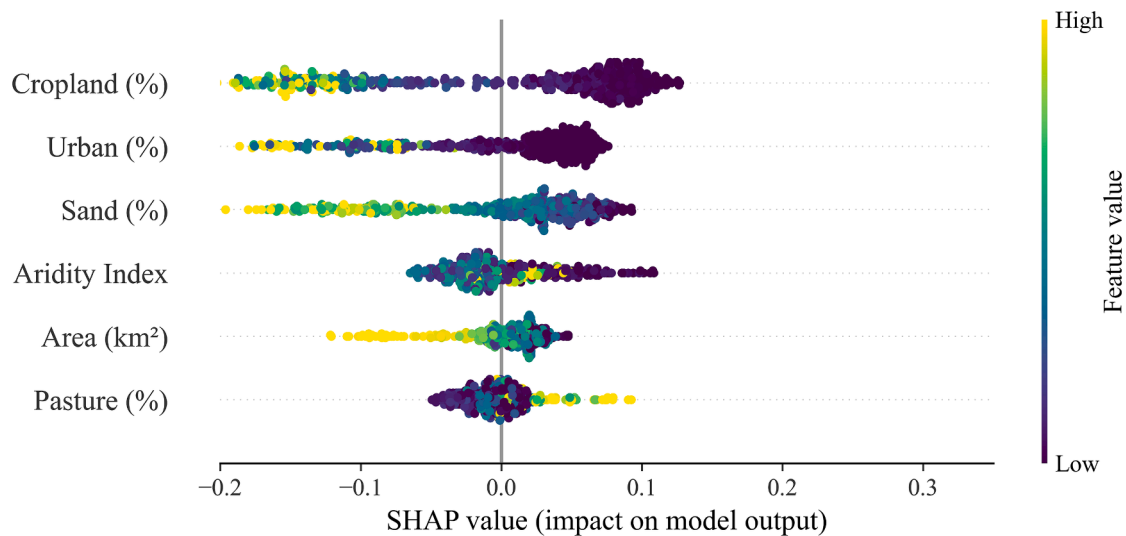


Fig. 8. Summary plot of SHAP values illustrating the impact of different features on the initial abstraction ratio for Group (1) (positive class $\lambda = 0.2$) in unbalanced dataset. Each point on the plot represents the SHAP value for a feature for a single prediction, indicating the contribution of that feature to moving the model output from the base value. The color scale from green to purple represents feature values from high to low, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a trade-off between Recall and Precision when balancing the dataset.

The SHAP graphs (Figs. 7 and 8) visualize the impact of attributes on the determination of an optimal λ (positive variable $\lambda = 0.2$), for both balanced and unbalanced datasets. Although the RFM highlights the importance of these attributes, it does not reveal their impact on individual observations. This nuance is captured by the SHAP plot, which displays a relatively balanced contribution of attributes to both datasets. The wide dispersion of the SHAP values along the horizontal axis, which resulted from the application of SMOTE-Tomek, showed that the distinction between classes was more visible for the balanced dataset. The attributes contributed similarly to the balanced dataset (Fig. 7), with Urban (%), cropland (%), and sand (%) showing a greater average impact on λ . A similar trend of balance was observed in the imbalanced dataset (Fig. 8), with cropland (%), Urban (%), and sand (%) also exhibiting a greater average impact on λ . The results found through SHAP values regarding average impact on predictions align with the inferences obtained from the RFA. The application of balancing clarified the model's generalization in underrepresented classes, the tendency of how an increase or decrease in the variable is associated with an output. The differences in the SHAP value graphs showed that balancing had a significant effect on the impact of each characteristic but the general trend observed before balancing remained consistent.

The results indicate that increases in pasture area (%) and drainage area correlated with the increase in the probability of λ to be 0.2, while their reduction increases the probability of λ to be 0.05. In contrast, an increase in cropland area (%) and urban area (%) were linked to a greater probability for $\lambda = 0.05$. Increased aridity index values typically arid regions, were associated with an increasing probability for $\lambda = 0.05$. Reductions in soil sand contents tended to increase the probability for $\lambda = 0.2$, while increases in sand contents favored towards $\lambda = 0.05$.

Throughout this analysis, it was observed a significant class imbalance for Group (2), with the majority class ($\lambda = 0.05$) representing 96.17%. In an attempt, the model showed high Accuracy (98.52%), but this was combined with zeros for Precision, Recall, and F1-score, suggesting a failure to predict the minority class accurately. The SMOTE-Tomek was not pursued further, considering that the generation of an excessive number of synthetic data in a small dataset could lead to overfitting problems, and the synthetic data might not accurately represent the original distribution.

4. Discussion

4.1. Performance of the CN calibration methods

The results indicated the LS as the best method regardless of the precipitation group (Table 1 and Fig. 4), surpassing the original NEH method. This outcome is aligned with previous studies (Ayers and Ramirez-Avila., 2024; Moglen et al., 2022; Pezan, 2022; Durán-Barroso et al., 2017; Oliveira et al., 2016; Ajmal et al., 2015; Lal et al., 2017) and contradicts Schneider and McCuen (2005), which claimed that the NEH method is superior. From the perspective of central tendency in CN values, the higher CN values associated with the NEH method (Fig. 3) have been noted in other studies (Ponce and Hawkins, 1996; Hawkins et al., 2009; Tedela et al., 2012; Oliveira et al., 2016; Junior Valle et al., 2019; Hawkins et al., 2020), suggesting that the tabulated CN values may be overestimated. When comparing the tabulated values in TR-55 (USDA, 1986) for urban areas, croplands, forests, and pastures with our results (S3 Figs. S13 to S18 and Tables S2 to S9), we found that the values obtained using the NEH method are close to the tabulated values. Moreover, runoff displays sensitivity to variations in land cover, vegetation types, agricultural practices in croplands, and the complex dynamics within forest ecosystems (Schwambach et al., 2024; He et al., 2023). This sensitivity reflects the variation in CN values and spatial variability (S4 Figs. S19 and S20) found in the results, dependent on the climatic and catchment characteristics of the watersheds across the countries. Furthermore, studies indicate that the CN method is not suitable for estimating runoff under various land covers, particularly in forested areas and specific types of croplands, while proving reasonable for pasture areas (Junior Valle et al., 2019; Oliveira et al., 2016; Tedela et al., 2012; Ponce and Hawkins, 1996). The NEH method, clearly resulted in biased CN estimations for Group (2), suggesting a high sensitivity to the rainfall sample used for calibration (Fig. 3c and Fig. 3d). The NEH method sensitivity has been addressed by other researchers (Moglen et al., 2022; Lian et al., 2020; Wilson et al., 2017), reinforcing the recommendation from Hawkins et al. (2009) regarding data censoring, and reducing the λ value showed a tendency to correct this bias. The sensitivity of the calibration stems from the NRCS-CN method's formulation, which accounts for the association between precipitation and runoff. Less precipitation decreases soil storage (S), which increases CN, whereas more rainfall increases S and decreases CN. Since lower-intensity precipitation events occur more frequently, using

the median method can lead to higher CN values, as these smaller events are more represented. Furthermore, the ASY method is not recommended for arid region watersheds (Farran and Elfeki, 2020), which contributes to the prominence of the LS method over the others.

Our findings also suggest a predominance of complacent-violent-inactive behaviors in the analyzed watersheds, reflecting the failure to capture the standard behavior. This was evidenced by the limited percentage of cases that demonstrated satisfactory adjustment for Group (1) (38.52% of cases for $\lambda = 0.2$ and 18.80% for $\lambda = 0.05$). However, the situation was the opposite for Group (2), (82.08% of cases for $\lambda = 0.2$ and 98.46% for $\lambda = 0.05$). Furthermore, an unsatisfactory performance was found when adopting the CN_{∞} , as shown in Table 1. For these watersheds the NRCS-CN is not the appropriate method and the runoff (Q) is better described as linearly dependent on rainfall (P) ($Q = CP$), where C is the runoff coefficient, which ranges from 0 to 1 (Hawkins et al., 2009). We found C values ranging from 0.00 to 0.76 (Group (1) and 0.00 to 0.57 (Group (2) in the present study (refer to S5 Figs. S21 and S22). Other authors have reported much-narrowed values for C, with the ranges of 0.05–0.276 (Pezan, 2022), 0.005–0.05 (Hawkins et al., 2020), 0.005–0.070 (Hawkins, 1973), 0.001–0.030 (Oliveira et al., 2016), 0.003–0.07 (Hawkins et al., 2015b), and 0.11–0.93 (Durbude et al., 2011).

Differently from previous studies, our analysis spans a wide variety of watersheds across a great variety of regions, which explains the contrasting ranges reported by the other studies. There is an association between complacent-violent-inactive behaviors in watersheds with predominant forest cover and high infiltration capacity (Hawkins et al., 2020), but complacent behavior is also observed in urban areas (Ling et al., 2020; Hawkins et al., 2009). Therefore, the NRCS-CN method has limitations when applied in forested areas, types of croplands and is reasonable for pasture areas (Junior Valle et al., 2019; Oliveira et al., 2016; Tedela et al., 2012; Ponce and Hawkins, 1996). Since a large percentage of watersheds in this study dataset represent rural areas, with predominant forest and cropland covers, they have a high infiltration capacity explaining the proportion of unsatisfactory metrics (Fig. 4). Subsurface stormflow may dominate watersheds with these characteristics (Bartlett et al., 2016). On the other hand, watersheds with more than 45% impermeable area have performed better with traditional methods (Simple and Rational) compared to the NRCS-CN method (Simpson et al., 2023). Soulis and Valiantzas (2012) observed that ASY CN values traditionally focused on estimating greater depths, resulting in flawed descriptions of the watershed response to low precipitation events. Therefore, the ASY method is not recommended for arid region watersheds (Farran and Elfeki, 2020). The predominance of compliant behavior found here highlights the importance of investigating the generalization of the method for all countries, as shown here and in other studies (Galbetti et al., 2022; Junior Valle et al., 2019; D'Asaro et al., 2014; Sartori et al., 2011).

4.2. Effects of initial abstraction ratio on the method performance

Building on the discussion of the λ value, our research supports the selection of $\lambda = 0.05$ for all methods (Table 1 and Fig. 4, Fig. S7, Tables S10 to S12), with the association of the LS method for Group (1) is to be more advisable. The preference for $\lambda = 0.05$, corroborates with a long list of authors who, over the last 20 years in different regions of the world, have pointed out that the value of 0.2 tends to overestimate runoff (Ramirez-Avila et al., 2024; Simpson et al., 2023; Pezan, 2022; Galbetti et al., 2022; Farran and Elfeki, 2020; Jiao et al., 2019; Alves et al., 2019; Junior Valle et al., 2019; Durán-Barroso et al., 2017; Lal et al., 2017; D'Asaro and Grillone, 2012; Shi et al., 2009; Woodward et al., 2003; Jiang, 2001). Another fact supporting the value of 0.05 is the outcome of the Wilcoxon test, as for Group (1) the estimated values were statistically similar to the observed runoff values, with 69.01% of the watersheds using $\lambda = 0.05$ exhibiting similar performance, while using $\lambda = 0.2$ only 59.13% exhibited similarity between simulated and

observed runoff. For both λ values for Group (2), the predicted results were statistically different from the observed values, with 100% of the watersheds showing. This difference may be attributed to the method's sensitivity to the onset of precipitation events, as well as pre-existing soil conditions (USDA-NRCS, 2004). The statistical test applied was intended to examine whether selecting $\lambda = 0.2$ or $\lambda = 0.05$ would result in different simulation outcomes. The results demonstrated that the choice of λ significantly affects the statistical similarity between the simulated and observed runoff values, highlighting the critical importance of λ selection in achieving an accurate method.

4.3. Rainfall-runoff events as a source of uncertainty

Besides our effort in adopting a reliable methodology, we understand that there is one major open uncertainty regarding selecting representative precipitation events and surface runoff used in estimating the methods' parameters (Durán-Barroso et al., 2017). Hawkins et al. (2009) recommends a threshold of 25.4 mm; however, this criterion has not been reevaluated (Durán-Barroso et al., 2017). The results indicate that applying a limit for $P > 25.4$ mm is more appropriate than $P > 0$ mm, as including all events leads to statistically significant differences and a lower proportion of watersheds with satisfactory performance. For events above this threshold, the median CNs were generally closer among the different methods compared to using all precipitation events. This indicated that the methods converge towards a more stable CN estimate for larger and more significant precipitation events. Censoring leads to a reduction of the bias caused by smaller precipitation events which increases the CN value (D'Asaro and Grillone, 2012; Shi et al., 2009), and avoids that a dataset with larger precipitation events increases the likelihood of Ia being insignificant (Lal et al., 2017), for which it is intrinsic to the NRCS-CN method. On the other hand, there is a variation of λ as a function of rainfall (Krajewski et al., 2020), and our results corroborate with the results found by Fu et al. (2011), suggesting that $\lambda = 0.05$ is more appropriate for events smaller than 25.4 mm. Durán-Barroso et al. (2017) identified thresholds of 14 mm and 57 mm in 11 U.S. watersheds, while Simpson et al. (2023) and Shi et al. (2009) found thresholds of 12.5 mm and 11.2 mm, respectively. Assaye et al. (2021) considered events of 10 mm because few exceeded 25 mm in Ethiopia. (Lal et al., 2017) suggested a value of 15 mm, considering it an arbitrary criterion. (Fu et al., 2011) also indicated that about 50% of the events were smaller than 25 mm but contributed to 20% of the total runoff volume in their study, making them necessary for accurately predicting runoff depth. Our results support the use of 25.4 mm as a threshold for representative events as suggested by (Durán-Barroso et al., 2017); however, similar to their findings, we highlight that its widespread application can lead to significant discrepancies, omitting important P-Q events. We suggest an intermediate value for areas less prone to heavy rainfall or with a higher likelihood of runoff, emphasizing the need to include smaller events to cover the diversity of hydrological processes.

4.4. Conversion of CN from $\lambda = 0.2$ to $\lambda = 0.05$

The equations developed in our study to calculate $S_{0.05}$ from $S_{0.2}$ (Equations (17) to (20)) had similarity between the groups of precipitation and demonstrated that the power model better captures the relationship between $S_{0.05}$ and $S_{0.2}$ compared to simple linearity, which aligns with previous studies (Hawkins et al., 2020; Woodward et al., 2003). Jiang (2001), Woodward et al. (2003) proposed $S_{0.05} = 1.33S_{0.2}^{1.15}$ (inches) and more recently Hawkins et al. (2020) proposed $S_{0.05} = 1.32S_{0.2}^{1.089}$ (inches) and with transformation $S_{0.05} = 0.8187S_{0.2}^{1.15}$ (mm), using a sample of 307 watersheds from the United States. This equation, derived for LS fits of CN and S to natural datasets P-Q, is in the draft document submitted by the ASCE-EWRI Curve Number Hydrology Committee, currently under review in the USDA-NRCS (USDA, 2017).

Our results, Equation (17) $S_{0.05} = 0.919S_{0.2}^{1.174}$, Group (1), and Equation (18) $S_{0.05} = 0.917S_{0.2}^{1.170}$, for Group (2), are close to those found by these authors but use a significantly larger sample (3578) (Fig. S27 in S8 for complements the comparison), capturing a wider range of regional characteristics for watersheds worldwide. For instance, when considering only the data from the United States, the equations $S_{0.05} = 1.17S_{0.2}^{1.114}$ ($n = 1260$, Group (1) and $S_{0.05} = 1.25S_{0.2}^{1.098}$ ($n = 1358$, Group (2)) were found (Table 2). The dataset used to develop these equations includes a greater number of watersheds with distinct characteristics that represent various regions than the previous authors.

Group (1), which represents the most common approach and methodology similar to that used by Jiang (2001), Woodward et al. (2003) and Hawkins et al. (2020), presented satisfactory metrics and greater alignment with these studies. These results indicate that while there are regional variations in watershed behavior (Table 2), the underlying relationship between $S_{0.2}$ and $S_{0.05}$ remains consistent across different hydrological conditions. Despite the study is focus on the practical applications, purely geographical divisions may not fully capture the hydrological variability among watersheds. As observed in cases like France and Germany (Table 2), regional watershed characteristics and sample sizes influence the parameters used in conversion equations. Therefore, future studies should consider hydrologically similar watershed groupings rather than strictly geographical boundaries when developing regional conversion equations.

Our findings highlight the importance of these regression equations for converting $S_{0.2}$ to $S_{0.05}$, seeking to provide more accurate and representative results. The large sample size and diversity of watersheds in our study enhance the robustness and generalizability of our equations, making them more suitable for broader applications compared to previous studies that used limited data. This is particularly significant because many studies proposing conversion equations used data from only one or a few watersheds (e.g., Jiang 2001; Woodward et al. 2003; Durán-Barroso et al., 2017; Junior Valle et al. 2019; Chin 2023b), which often focus on regions with specific characteristics. In contrast, our study encompasses a wide range of watershed characteristics across different regions.

The change in λ (to $\lambda = 0.05$) has a significant impact on hydrological modeling and water resource design worldwide (Moglen et al., 2022, 2023). The extensive data used in our study overpasses conversion problems that previous equations in the literature were unable to cover due to the concentration on local studies and low representativeness. These regression equations to convert $S_{0.2}$ to $S_{0.05}$, and consequently $CN_{0.05}$, for the entire dataset and each of the 32 countries in the study, are crucial for future applications of the NRCS-CN method considering $\lambda = 0.05$. These equations update the previously tabled CN values in TR-55 (USDA-NRCS, 1986), making our equations highly relevant for diverse projects and studies linked to society.

4.5. Influence of climatic and catchment characteristics on CN and λ

Our results indicate that land cover is the most influential factor in determining the optimal λ value, more so than climatic characteristics, with cropland exerting the greatest impact and drainage area the least. The aridity index, drainage area, soil sand content, and pasture area contribute to λ in a slightly lesser yet well-balanced manner. Larger watersheds and arid regions tend to favor $\lambda = 0.05$, while pasture-dominated areas are more associated with $\lambda = 0.2$. A more detailed discussion on the climatic and catchment characteristics influencing CN and λ is provided in the supporting information S9.

4.6. Perspectives and reasonableness of study

Since its inception, the NRCS-CN method has undergone significant transformations, and more recently, the correct value of λ has been debated. Many studies have pointed to the value of 0.05, but they were

mostly conducted with a limited number of watersheds and datasets. The present study, in contrast, stands out for analyzing a larger number of watersheds distributed across the five continents to assess this λ value, in addition to other parameters, based on observed data from these watersheds. Unlike previous studies, which were limited to a few metrics, our analysis employs five distinct metrics, addressing different possibilities of error since each has its advantages and disadvantages relative to the others. In addition to following traditionally employed analyses, we have turned to the latest data science tools, such as ML, to determine the appropriate value of λ . Therefore, our findings suggest the use of $\lambda = 0.05$ as a general parameter in cases where local calibration is not feasible, which is common in the practical application of the method in engineering. Extensive literature corroborates this value, indicating the necessity of converting CN values.

Our benchmark for the metrics is strict, considering only performances deemed good. Given the scale of the data, a less restrictive benchmark could be applied. For example, if the mean runoff is chosen as a benchmark, as in cases where $NSE > 0$ and $KGE > -0.41$ (Knoben et al., 2019), the central tendency of the results remains unchanged. Moreover, when comparing the groups using the same sample size (excluding the 180 watersheds with no rainfall events exceeding 25.4 mm), the results of the study did not show significant sensitivity.

Despite the importance of our findings, we highlight that our study has some inherent limitations related to the data and methods used that might affect the results reported here. For the method, we emphasize that we worked with daily rainfall-runoff events, which prevented us from calculating the observed λ for sub-hourly events, which would encompass the effects of rainfall duration and intensity. Furthermore, we considered the average antecedent moisture (II), which can lead to the smoothing of CN and Ia values. These antecedent moisture conditions affect these values in some regions. As pointed out by Lian et al. (2020), the CN value is the primary source of error in the method. Regarding the data, uncertainties may arise due to incompatibility along the catchment boundaries between different delineations of watersheds in different Caravan datasets (Kratzert et al., 2023), which would lead to uncertainties in precipitation data extracted from ERA5-Land (Muñoz Sabater et al., 2021). The runoff separation method, although recommended in literature, may not be suitable for all evaluated catchments.

The combination of ML techniques with hydrology, especially with explainable artificial intelligence (XAI) methods, has been increasingly applied in recent studies due to its ability to capture patterns not identifiable by traditional methods, generate hypotheses, identify which variables are most influential, and understand the relationships between variables (Nearing et al., 2024; Slater et al., 2024; Jiang et al., 2024a; Jiang et al., 2024b; Mamalakos et al., 2023). However, these techniques face criticism, such as being highly dependent on the selected data, the use of interpretability techniques for data-driven models rather than inherently interpretable models, and the challenge of distinguishing correlation from causality (Jiang et al., 2024b; Rudin, 2019; Breiman, 2001).

Many studies have mapped CN values at high global resolution (Wu et al. 2023; Sujud and Jaafar, 2022; Jaafar et al., 2019; Zeng et al., 2017), but these values may be associated with errors when using the original CN table, which has been shown not to be generalized to all countries. Our work will be useful in improving these products. For future research, we suggest including the comparison of these values, addressing soil moisture, rainfall intensity, ranking pairs, incorporating other ML methods.

5. Conclusion

In this study, we computed the CN values and investigated the effects of initial abstraction ratio on the performance of runoff estimation for 3578 watersheds distributed across five continents. In addition, we explored the main relationships of watershed characteristics with λ and CN values. We investigated two values of λ (0.05 and 0.02) and

computed CN values from daily rainfall and runoff data using three methods: Asymptotic (ASY), Median (NEH), and Least-Squares (LS). Our findings provide insights into the method's parameters, the evaluation of precipitation thresholds, the association of physical characteristics with CN values, the most suitable λ value, and the physical characteristics that influence it, as well as the quantitative relationship between them. These insights include:

- The Least-Squares (LS) method stood out with the best accuracy in estimating CN values, followed by the asymptotic (ASY) method.
- The use of $\lambda = 0.05$ yields more suitable runoff estimations than when using the standard value of $\lambda = 0.2$.
- Similar to λ , the precipitation threshold for calibrating CN values depends on local conditions, but generally, the application of a 25.4 mm threshold stands as the best option to be generalized without penalizing the applicability of the NRCS-CN method.
- High CN values, indicative of a high runoff potential, are governed by the variability in slope, precipitation, and silt content in the soil, while low CN values are associated with forest cover alongside this cover correlated with regions characterized by sandy soils, and this association extends to the aridity index.
- The most influencing characteristic for determining an optimal λ value is land cover, with cropland area having the most influence, while the drainage area has the least. The aridity index, soil sand content, urban area, and pasture area also influence the decision on λ to a slightly lesser extent, yet in a balanced manner.
- Overall, this study found that a general λ value of 0.05 is more suitable for runoff estimations than the adopted standard of 0.2. Given the high variability of hydrological processes across watersheds worldwide, watersheds with higher percentages of pasture tend to use $\lambda = 0.2$, while those with higher agricultural area, soil sand content, watershed area, and sparsely built urban areas are better suited to $\lambda = 0.05$. Low aridity index regions are associated with $\lambda = 0.2$, while the opposite conditions favor $\lambda = 0.05$.

This study underscores the significance of hotspots in the NRCS-CN method parameters as emergent from combined hydrological processes rather than linear relationships. Therefore, varying characteristics contribute distinctly, leading to patterns that cannot be generalized by the independent contribution of certain key factors worldwide. However, for the practical application, this research presents highlights to make the method more accurate, providing equations considering $\lambda = 0.05$, which can be used to update the previously tabled CN values in TR-55. This influences various disciplines where it is applied, including hydrological processes, sediment management, flood forecasting, watershed modeling water resources engineering, and others.

CRedit authorship contribution statement

Abderraman R. Amorim Brandão: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Visualization, Writing – original draft. **Dimaghi Schwamback:** Methodology, Visualization, Writing – review & editing. **André Simões Ballarin:** Methodology, Visualization. **John J. Ramirez-Avila:** Methodology, Supervision, Visualization, Writing – review & editing. **José Goes Vasconcelos Neto:** Writing – review & editing. **Paulo Tarso S. Oliveira:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2025.132941>.

Data availability

All data and code from Brandão et al. (2024) are available via an open-access repository at <https://doi.org/10.5281/zenodo.12734750>.

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